Hongjie Fang

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Education

Shanghai Jiao Tong University *2022/09 – 2027/06 (expected)*
Ph.D. Candidate, Wu Wenjun Honorable Class, Computer Science Shanghai, China
Advisor: Cewu Lu.

Shanghai Jiao Tong University *2018/09 – 2022/06*
Bachelor of Engineering, major in Computer Science and Engineering Shanghai, China
Bachelor of Economics, minor in Finance
- GPA 4.03 / 4.3, Ranking: 2 / 149.

Research Interest

My research interests mainly lie on the **robotics** fields, specifically, robotic manipulation (contact-rich manipulation, dexterous manipulation), robot learning (imitation learning, reinforcement learning), data collection, and grasping.

Goal: To enable robots to perform various tasks in the real world under any circumstances with human-level dexterity.

Awards

Top Reviewer, NeurIPS 2026 Evaluations and Datasets Track *2026*
Best Paper Award, ICRA 2024 *2024*
Best Paper Award, IROS 2023 Manipulation and Grasping Workshop *2023*
Shanghai Outstanding Graduates *2022*
Shanghai Scholarship *2020*
National Olympics in Informatics (**NOI**) *Bronze Medal* *2017*
National Olympics in Informatics in Provinces (**NOIP**) *First Prize* *2015 & 2016*

Publications & Preprints **Total Citations: 5518**

* denotes equal contribution, # denotes corresponding author(s).

1. FP2: Equipping Robotic Foundation Models with Force Control
Hongjie Fang*, Shirun Tang*, Junjian Hu, Shidong Zhang, Derek Zhang, Linhao Chen, Dehai Li, Mingyu Mei, Wanxi Liu, Cewu Lu, and Shiquan Wang#
arXiv preprint, 2026.

2. Rethinking Visual Embodiment Dependence in Visuomotor Policies
Hongjie Fang, Yuxuan Lu, Chenxi Wang, Haoxiang Qin, Shirun Tang, Zihao He, Shangning Xia, Jingjing Chen, Wanxi Liu, Shiquan Wang#, Cewu Lu#
arXiv preprint, 2026.

3. Force Policy: Learning Hybrid Force-Position Control Policy under Interaction Frame for Contact-Rich Manipulation
Hongjie Fang*, Shirun Tang*, Mingyu Mei*, Haoxiang Qin, Zihao He, Jingjing Chen, Ying Feng, Chenxi Wang, Wanxi Liu, Zaixing He, Cewu Lu, Shiquan Wang#
*Robotics: Science and Systems (**RSS**), 2026.*

4. AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons
Hongjie Fang*, Chenxi Wang*, Yiming Wang*, Jingjing Chen*, Shangning Xia, Jun Lv, Zihao He, Xiyan Yi, Yunhan Guo, Xinyu Zhan, Lixin Yang, Weiming Wang, Cewu Lu#, Hao-Shu Fang#
*Conference on Robot Learning (**CoRL**), 2025, **(Oral Presentation)**.*

5. AirExo: Low-Cost Exoskeletons for Learning Whole-Arm Manipulation in the Wild
Hongjie Fang*, Hao-Shu Fang*, Yiming Wang*, Jieji Ren, Jingjing Chen, Ruo Zhang, Weiming Wang, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024.*

6. TransCG: A Large-Scale Real-World Dataset for Transparent Object Depth Completion and A Grasping Baseline
Hongjie Fang, Hao-Shu Fang, Sheng Xu, Cewu Lu#
*IEEE Robotics and Automation Letters (**RA-L**), 2022; Presented at **ICRA** 2023.*

7. History-Aware Visuomotor Policy Learning via Point Tracking
Jingjing Chen*, **Hongjie Fang***, Chenxi Wang, Shiquan Wang#, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2026.*

8. FoAR: Force-Aware Reactive Policy for Contact-Rich Robotic Manipulation
Zihao He*, **Hongjie Fang***, Jingjing Chen, Hao-Shu Fang#, Cewu Lu#
*IEEE Robotics and Automation Letters (**RA-L**), 2025; Presented at **IROS** 2025.*

9. Learning Dexterous Manipulation with Quantized Hand State
Ying Feng*, **Hongjie Fang***#, Yinong He*, Jingjing Chen, Chenxi Wang, Zihao He, Ruonan Liu, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2026.*

10. AnyDexRT: Calibration-Free Dexterous Hand Retargeting with Few-Shot Human Guidance
Chenxi Wang*, Ying Feng*, **Hongjie Fang#**, Shangning Xia, Lixin Yang, Chuan Wen, Cewu Lu#
*Conference on Robot Learning (**CoRL**), 2026.*

11. Asynchronous Multimodal Diffusion Policy Composition via Latency-Aware Guidance Fusion
Zihao He*, **Hongjie Fang***, Shirun Tang, Cewu Lu#, Hao-Shu Fang#
arXiv preprint, 2026 [RSS 2026 Workshop on Diffusion for Robot Learning].

12. RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot
Hao-Shu Fang, **Hongjie Fang**, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, Haoyi Zhu, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024*

13. RISE: 3D Perception Makes Real-World Robot Imitation Simple and Effective
Chenxi Wang, **Hongjie Fang**, Hao-Shu Fang#, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2024.*

14. CAGE: Causal Attention Enables Data-Efficient Generalizable Robotic Manipulation
Shangning Xia, **Hongjie Fang**, Cewu Lu#, Hao-Shu Fang#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2025.*

15. Towards Effective Utilization of Mixed-Quality Demonstrations in Robotic Manipulation via Segment-Level Selection and Optimization
Jingjing Chen, **Hongjie Fang**, Hao-Shu Fang, Cewu Lu#
*IEEE International Conference on Robotics and Automation (**ICRA**), 2025.*

16. X-Imitator: Spatial-Aware Imitation Learning via Bidirectional Action-Pose Interaction
Kai Xiong, **Hongjie Fang**, Lixin Yang, Cewu Lu#
arXiv preprint, 2026.

17. AnyGrasp: Robust and Efficient Grasp Perception in Spatial and Temporal Domains
Hao-Shu Fang, Chenxi Wang, **Hongjie Fang**, Minghao Gou, Jirong Liu, Hengxu Yan, Wenhai Liu, Yichen Xie, Cewu Lu#
*IEEE Transaction on Robotics (**T-RO**), 2023; Presented at **ICRA** 2024, **(Best Paper Award on IROS 2023 Workshop)**.*

18. Motion Before Action: Diffusing Object Motion as Manipulation Condition
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE Robotics and Automation Letters (**RA-L**), 2025; Presented at **ICRA** 2026.*

19. Dense Policy: Bidirectional Autoregressive Learning of Actions
Yue Su*, Xinyu Zhan*, **Hongjie Fang**, Han Xue, Hao-Shu Fang, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE/CVF International Conference on Computer Vision (**ICCV**), 2025.*

20. Knowledge-Driven Imitation Learning: Enabling Generalization Across Diverse Conditions
Zhuochen Miao*, Jun Lv*, **Hongjie Fang**, Yang Jin, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.*

21. Open X-Embodiment: Robotic Learning Datasets and RT-X Models
Open X-Embodiment Collaboration, 147 authors
*IEEE International Conference on Robotics and Automation (**ICRA**), 2024, **(Best Paper Award)**.*

22. Flexible Handover with Real-Time Robust Dynamic Grasp Trajectory Generation
Gu Zhang, Hao-Shu Fang, **Hongjie Fang**, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2023.*

23. AnyDexGrasp: Learning General Dexterous Grasping for Any Hands with Human-level Learning Efficiency
Hao-Shu Fang#, Hengxu Yan, Zhenyu Tang, **Hongjie Fang**, Chenxi Wang, Cewu Lu#
arXiv preprint, 2025 [ICLR 2025 Workshop on Robot Learning].

24. ChronoFlow Policy: Unifying Past-Future Interaction Flow in Visuomotor Policy Learning
Bokai Lin*, Yifu Xu*, Xinyu Zhan, **Hongjie Fang**, Jialin Tian, Fu-Cheng Zhang, Yong-Lu Li, Cewu Lu, Lixin Yang#
*European Conference on Computer Vision (**ECCV**), 2026.*

25. LIDEA: Human-to-Robot Imitation Learning via Implicit Feature Distillation and Explicit Geometry Alignment
Yifu Xu*, Bokai Lin*, Xinyu Zhan, **Hongjie Fang**, Yong-Lu Li, Cewu Lu, Lixin Yang#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2026.*

26. SIME: Enhancing Policy Self-Improvement with Modal-Level Exploration
Yang Jin*, Jun Lv*, Wenye Yu, **Hongjie Fang**, Yong-Lu Li, Cewu Lu#
*IEEE/RSJ International Conference on Intelligent Robots and Systems (**IROS**), 2025.*

27. Graspness Discovery in Clutters for Fast and Accurate Grasp Detection
Chenxi Wang*, Hao-Shu Fang*, Minghao Gou, **Hongjie Fang**, Jin Gao, Cewu Lu#
*IEEE/CVF International Conference on Computer Vision (**ICCV**), 2021.*

28. Target-Referenced Reactive Grasping for Dynamic Objects
Jirong Liu, Ruo Zhang, Hao-Shu Fang, Minghao Gou, **Hongjie Fang**, Chenxi Wang, Sheng Xu, Hengxu Yan, Cewu Lu#
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**), 2023.*

29. HINT: Human-Intent Inception for Long-Horizon Robot Manipulation
Mingyu Mei, Haojie Xu, Shihao Jin, Zibo Dai, Qihao Cheng, Zhengrui Lv, **Hongjie Fang**, Shirun Tang, Guang Chen, Xinyue Zhao, Huiliang Shen, Zaixing He#
arXiv preprint, 2026.

Academic Services

Associate Editor for conference *ICRA*.
Reviewer for journal *T-RO*, *RA-L*, *T-CYB*, *T-MECH*, *T-ASE*.
Reviewer for conferences including *ICRA*, *IROS*, *RSS*, *CoRL*, *NeurIPS*, *ICLR*, *CVPR*, *ECCV*, *etc.*

Invited Talks

- 03/2024, **Echo AI @ USyd**, *Towards Efficient Robot Imitation Learning from Human Demonstrations.*
- 11/2024, **Zhixingxing**, *Towards Efficient Robot Imitation Learning from Human Demonstrations.*
- 03/2025, **Yang Gao Group @ THU**, *Towards Generalizable Imitation Learning from Human Demonstrations.*
- 08/2025, **3D CVer**, *Towards Generalizable Imitation Learning from Human Demonstrations.*
- 08/2025, **Sharpa**, *Towards Generalizable Imitation via Scalable Data and Robust Policy.*
- 09/2025, **Galbot**, *Building Capable and Generalizable Policies for Robotic Manipulation.*
- 09/2025, **Shenlanxueyuan**, *Towards Generalizable Imitation via Scalable Data and Robust Policy.*
- 10/2025, **The Robot Dexterity Lab @ Duke**, *AirExo-2: Scaling up Generalizable Robotic Imitation Learning with Low-Cost Exoskeletons.*
- 09/2026, **ECUST**, Imitation Learning for Manipulation: Past, Present and Future.

Teaching

Teaching Assistant, *Algorithm and Complexity* — *Spring, 2022*
Teaching Assistant, *C++ Programming Language (Honor)* — *Fall, 2020*
Teaching Assistant, *Data Structure (Honor)* — *Spring, 2019*

Teaching Assistant, *Linear Algebra (Honor)* *Fall 2021 & Fall, 2022*
Teaching Assistant, *Mathematical Analysis (Honor)* *Fall, 2020 & Fall, 2021*